

The Valid Dictionary: Newtonian Mechanics

Can Valid

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Valid Dictionary (VD) Document: Newtonian Mechanics as a Valid Dictionary	Status:
DRAFT	
Internal ID: vd-newton	Project tag: vd-project

Point. Present a complete, minimal Newtonian mechanics encoding as a VD instance, demonstrating auditability and explicit structure.

Target goals.

- Provide a finished Newtonian VD with clear entry indexing and cross-references.
- Encode the relevant Newtonian “laws” as VD triplets consistent with **vd-core**.
- Keep commentary minimal: enough to read and verify the dictionary, not to persuade.

Non-goals.

- Arguing why the case study is important (belongs in **vd-newton-analysis**).
- Introducing new VD rules or changing the core spec.
- Covering advanced Newtonian topics beyond “basics” needed for demonstration.

Target audience. Readers who want to see VD used on real physics; skeptics who want something concrete to inspect.

Dependencies / references. Depends on **vd-core** for definitions, notation, and validity criteria.

1 Design 1

1.1 Non-law kinematics scaffold

E1 time := “A real parameter $t \in \mathbb{R}$ used to order events; differences Δt are durations measured by a clock.”

E2 reference_frame := “A convention for assigning spatial coordinates and a time coordinate t to events: an origin, axes, and a clock.”

E3 position := “In a reference_frame, an object’s location at time t represented by a vector $\mathbf{r}(t) \in \mathbb{R}^3$.”

E4 displacement := “ $\Delta \mathbf{r} = \mathbf{r}(t_2) - \mathbf{r}(t_1)$.”

E5 trajectory := “The time-ordered curve $\{\mathbf{r}(t) : t \in I\}$ for some interval $I \subset \mathbb{R}$.”

E6 velocity := “ $\mathbf{v}(t) = d\mathbf{r}/dt$.”

E7 speed := “ $v(t) = \|\mathbf{v}(t)\|$.”

E8 acceleration := “ $\mathbf{a}(t) = d\mathbf{v}/dt = d^2\mathbf{r}/dt^2$.”

E9 path_length := “From t_1 to t_2 : $s = \int_{t_1}^{t_2} \|\mathbf{v}(t)\| dt$.”

E10 point_particle := “An idealized object whose size and internal structure are neglected; its motion in a reference_frame is fully represented by a trajectory $\mathbf{r}(t)$.”

E11 straight_line := “A set $\mathbf{r}(\lambda) = \mathbf{r}_0 + \lambda \mathbf{u}$ with fixed $\mathbf{r}_0$, nonzero direction $\mathbf{u}$, and $\lambda \in \mathbb{R}$.”

1.2 Uniform motion intermediates (non-law)

- E12** `uniform_motion_on_interval(I)` := “A point_particle is in `uniform_motion_on_interval(I)` if its velocity is the same at every time in I ; i.e. $\forall t_1, t_2 \in I, \mathbf{v}(t_1) = \mathbf{v}(t_2)$.”
- E13** `uniform_motion_at_time(t_0)` := “A point_particle is in `uniform_motion_at_time(t_0)` if $\exists \varepsilon > 0$ such that $\forall t \in (t_0 - \varepsilon, t_0 + \varepsilon), \mathbf{v}(t) = \mathbf{v}(t_0)$.”

1.3 Newton I (triplet: uniform motion as bridge; no acceleration)

- E14** `uniform_motion_on_interval(I)` := “The state of motion exhibited by a free_particle when described in an inertial_frame over the interval I .”
- E15** `inertial_frame` := “A reference_frame in which every free_particle exhibits `uniform_motion_at_time(t)` (equivalently: `uniform_motion_on_interval(I)` on any interval of free motion).”
- E16** `free_particle` := “A point_particle that, when described in an inertial_frame, exhibits `uniform_motion_at_time(t)` (equivalently: `uniform_motion_on_interval(I)`) whenever treated as free.”

1.4 Newton II Law

- E17** `acceleration` := “For a point_particle described in an inertial_frame, acceleration equals net_force divided by inertial_mass: $\mathbf{a} = \mathbf{F}_{\text{net}} / \text{inertial_mass}$ (equivalently: $\mathbf{F}_{\text{net}} = \text{inertial_mass} \cdot \mathbf{a}$).”
- E18** `net_force` := “The vector quantity satisfying `net_force = inertial_mass · acceleration` for the point_particle (in an inertial_frame).”
- E19** `inertial_mass` := “The positive scalar coefficient m such that `net_force = m · acceleration` for the point_particle (in an inertial_frame).”

1.5 Force sum Law

- E20** `net_force(p, t)` := “For any particle p and time t , `net_force(p, t)` is the vector-sum of all acting_force elements in `interacting_forces_set(p, t)` (equivalently: `net_force(p, t) =  $\sum_{F \in \text{interacting\_forces\_set}(p, t)} F$`).”
- E21** `interacting_forces_set(p, t)` := “The set of acting_force vectors that the modeler judges to be acting on particle p at time t (i.e. the force-contributions currently included for p at t).”
- E22** `acting_force` := “A vector force-contribution F that appears as an element of `interacting_forces_set(p, t)` for some particle p and time t .”

1.6 Meta-typing (triplet: raw classes and recognition-based instantiation)

- E23** `raw_class` := “An abstract object-type.”
- E24** `raw_class` := “An abstract object-type specified by `class_specific_features(C)`; an object x is treated as `instance_of(x, C)` when it is recognized as possessing the `class_specific_features(C)`.”
- E25** `class_specific_features(C)` := “The set/list of features that specify raw_class C ; these features are whatever a modeler must recognize x as having in order to treat x as `instance_of(x, C)`.”
- E26** `instance_of(x, C)` := “ x is treated as an instance of raw_class C when x is recognized as possessing `class_specific_features(C)`.”

1.7 Action-Reaction Law(1st part of Newton III)

- E26** `Acting force.` := “*Acting force* is a force-contribution that is either a *canonical_force(acting_object)* or a *reaction_force(acting_object)*.”
- E28** `Canonical force.` := “A *canonical_force(acting_object)* is an acting force designated as canonical; the corresponding *reaction_force(acting_object)* is the acting force defined as the negative of it.”
- E29** `Reaction force.` := “A *reaction_force(acting_object)* is an acting force defined as the negative of *canonical_force(acting_object)*.”

1.8 Acting object Law

- E30** *acting_object* := An *acting_object* is an object equipped with both an *activation_condition(acting_object)* and a corresponding *canonical_force(acting_object)*. (This is a class)
- E31** *Activation condition.* := If a particle fulfills the *activation_condition(acting_object)* the *canonical_force(acting_object)* will be an acting-force on the particle.
- E32** *Canonical force (of an acting object).* := The *canonical_force(acting_object)* is the acting-force of a particle that fulfills the *activation_condition(acting_object)*.

1.9 Non-law

- E33** *paired_particle(acting_object)* := Each *acting_object* carries an associated particle, called its *paired_particle*. The acting object is considered to be located at the location of its paired particle. The particle may act as the object's COM.
- E34** *mechanical_system* := An arbitrary subset of particles $\{p_1, \dots, p_n\}$ selected by the observer for analysis.
- E35** *interaction_candidate* := For a particle p in a mechanical system S at time t , an *interaction_candidate* is any acting force in *interacting_forces_set*(p, t).

1.10 Mechanical Closure Law (2nd Part of Newton III)

- E36** *interaction_candidate* := An *interaction_candidate* is an elementary force-unit, every one of which must be able to be placed in an *interaction_pair* for the mechanical system to be a *mechanically closed system*.
- E37** *interaction_pair* := An *interaction_pair* is the form into which *interaction_candidates* are placed in a *mechanically closed system*.
- E38** *mechanically_closed_system* := A *mechanically closed system* is a *mechanical system* in which every *interaction_candidate* can be placed in an *interaction_pair*.

1.11 Non-law

- E39** *interaction_pair* := Two *interaction_candidates* form an *interaction_pair* when they correspond to the same *acting_object*, one being the *canonical_force* associated with that acting object and the other being the *reaction_force* associated with that acting object, acting on the object's *paired_particle*.
- E40** *mechanically_closed_system* := A *mechanically closed system* is a *mechanical system* whose future behaviour is determined by the equations produced from the interactions internal to it.

2 Design 2

2.1 Kinematics (Non-law)

The entries in this scaffold express standard kinematic notions from introductory classical mechanics, cross-checked against sources such as Kleppner–Kolenkow and Taylor, but rewritten to satisfy VD entry-log dependency order. In particular, the early non-law entries are intended to capture familiar pre-law notions—time, reference frame, point particle, position, displacement, velocity, speed, acceleration, trajectory, straight line, path length, relative position, and relative velocity—in a form that uses only residue and earlier headwords. The conceptual content is source-backed; the phrasing and ordering are ours.

The Newton I and Newton II blocks that follow should be read somewhat differently. Their underlying physics is standard Newtonian mechanics, but their presentation as chimney, triplet, and wall entries belongs to the VD architecture rather than to any single textbook. Thus the scaffold entries are the ones most directly defended by the cited mechanics sources, while the later law-units are best understood as VD-native reorganisations of standard material.

- E1 time** := A real parameter $t \in \mathbb{R}$ used to order events; differences $\Delta t = t_2 - t_1$ are durations.
- E2 reference_frame** := A convention for assigning spatial coordinates and a time coordinate to events: an origin, a set of basis vectors, and a clock.
- E3 point_particle** := An idealised object whose spatial extent and internal structure are neglected.
- E4 position** := The location of a point particle at time t in a reference frame, represented by a vector $r(t) \in \mathbb{R}^3$.
- E5 displacement** := The change in position: $\Delta r = r(t_2) - r(t_1)$.
- E6 velocity** := $v(t) = \frac{dr}{dt}$.
- E7 speed** := The scalar magnitude of velocity: $\text{speed}(t) = \|v(t)\|$.
- E8 acceleration** := $a(t) = \frac{dv}{dt} = \frac{d^2r}{dt^2}$.
- E9 trajectory** := The map $t \mapsto r(t)$ giving the positions of a point particle over some interval $I \subseteq \mathbb{R}$.
- E10 straight_line** := A set of points $\{r_0 + \lambda u : \lambda \in \mathbb{R}\}$ for fixed $r_0 \in \mathbb{R}^3$ and nonzero direction $u \in \mathbb{R}^3$.
- E11 path_length** := For a differentiable trajectory from t_1 to t_2 , the arc length is $s = \int_{t_1}^{t_2} \|v(t)\| dt$.
- E12 relative_position** := For two point particles in the same reference frame, the relative position is $r_{12}(t) = r_2(t) - r_1(t)$.
- E13 relative_velocity** := For two point particles in the same reference frame, the relative velocity is $v_{12}(t) = v_2(t) - v_1(t) = \frac{dr_{12}}{dt}$.
- E14 uniform_motion** := The state of being either stationary or moving along a straight line with constant speed. Equivalently, a point particle is in uniform motion on an interval I if its velocity is constant throughout I ; i.e. $\forall t_1, t_2 \in I, v(t_1) = v(t_2)$. A point particle is in uniform motion at time t_0 if $\exists \varepsilon > 0$ such that it is in uniform motion on $(t_0 - \varepsilon, t_0 + \varepsilon)$.

2.2 Newton's 1st Law

- E15 uniform_motion** := The state of motion exhibited by a free particle when described in an inertial frame.
- E16 inertial_frame** := A reference frame in which every free particle exhibits uniform motion.
- E17 free_particle** := A point particle that, when described in an inertial frame, exhibits uniform motion.

2.3 Non-law

- E18 inertial_frame** := A reference frame standard for Newtonian analysis of motion; historically, a Galilean reference frame.
- E19 free_particle** := A point particle not subject to external influence relevant to its motion.
- E20 inertial_acceleration** := The acceleration of a point particle as measured in an inertial frame.

2.4 Newton's 2nd Law

- E21 inertial_acceleration** := For a point particle, inertial acceleration equals net force divided by inertial mass: $a = F_{\text{net}}/m$.
- E22 net_force** := The vector quantity satisfying $F_{\text{net}} = m \cdot a$ for a point particle, where a is inertial acceleration.
- E23 inertial_mass** := The positive scalar coefficient m such that $F_{\text{net}} = m \cdot a$ for a point particle, where a is inertial acceleration.

2.5 Force sum Law

- E24** $\text{net_force} := \text{net_force}(p, t)$ is the vector sum of the *acting_force* elements in $\text{interacting_forces_set}(p, t)$.
- E25** $\text{interacting_forces_set} :=$ The set of *acting_force* elements taken to be acting on particle p at time t , whose vector sum is $\text{net_force}(p, t)$.
- E26** $\text{acting_force} :=$ A vector force-contribution that appears as an element of some $\text{interacting_forces_set}(p, t)$ and thereby contributes to $\text{net_force}(p, t)$.

2.6 Meta-typing (triplet: raw classes and recognition-based instantiation)

- E27** $\text{raw_class} :=$ An abstract object-type.
- E28** $\text{raw_class} :=$ An abstract object-type specified by $\text{class_specific_features}(C)$; an object x is treated as *instance_of*(x, C) when it is recognized as possessing the $\text{class_specific_features}(C)$.
- E29** $\text{class_specific_features}(C) :=$ The set of features that specify raw class C ; these features are whatever a modeler must recognize x as having in order to treat x as *instance_of*(x, C).
- E30** $\text{instance_of}(x, C) :=$ x is treated as an instance of raw class C when x is recognized as possessing $\text{class_specific_features}(C)$.

2.7 Action-Reaction Law (1st part of Newton III)

- E31** $\text{acting_force} :=$ An *acting_force* is a force-contribution that is either a $\text{canonical_force}(\text{acting_object})$ or a $\text{reaction_force}(\text{acting_object})$.
- E32** $\text{canonical_force}(\text{acting_object}) :=$ A $\text{canonical_force}(\text{acting_object})$ is an *acting_force* designated as canonical; the corresponding $\text{reaction_force}(\text{acting_object})$ is the *acting_force* defined as the negative of it.
- E33** $\text{reaction_force}(\text{acting_object}) :=$ A $\text{reaction_force}(\text{acting_object})$ is an *acting_force* defined as the negative of $\text{canonical_force}(\text{acting_object})$.

2.8 Acting object Law

- E34** $\text{acting_object} :=$ An *acting_object* is an object equipped with both an $\text{activation_condition}(\text{acting_object})$ and a corresponding $\text{canonical_force}(\text{acting_object})$.
- E35** $\text{activation_condition}(\text{acting_object}) :=$ If a *point_particle* fulfills the $\text{activation_condition}(\text{acting_object})$ of an *acting_object*, the $\text{canonical_force}(\text{acting_object})$ of that *acting_object* will be an *acting_force* on the *point_particle*.
- E36** $\text{canonical_force}(\text{acting_object}) :=$ The $\text{canonical_force}(\text{acting_object})$ is the *acting_force* that an *acting_object* exerts on a *point_particle* that fulfills the $\text{activation_condition}(\text{acting_object})$.

2.9 Non-law

- E37** $\text{mechanical_composition}(\text{point_particle}) :=$ The $\text{mechanical_composition}(\text{point_particle})$ of a *point_particle* is the set of *acting_objects* that the *point_particle* has. When a *point_particle* serves as the representative, or centre of mass, of a body, all *acting_objects* of that body are included in the $\text{mechanical_composition}(\text{point_particle})$ of that *point_particle*.
- E38** $\text{paired_particle}(\text{acting_object}) :=$ The $\text{paired_particle}(\text{acting_object})$ of an *acting_object* is the *point_particle* whose $\text{mechanical_composition}(\text{point_particle})$ includes that *acting_object*. The *acting_object* is modelled as located at the position of its $\text{paired_particle}(\text{acting_object})$.
- E39** $\text{mechanical_system} :=$ An arbitrary subset of particles selected by the observer for analysis.
- E40** $\text{interaction_candidate} :=$ For a particle p in a *mechanical_system* S at time t , an *interaction_candidate* is any *acting_force* in $\text{interacting_forces_set}(p, t)$.

2.10 Mechanical Closure Law (2nd Part of Newton III)

- E41** `interaction_candidate` := An *interaction_candidate* is an elementary force-unit, every one of which must be able to be placed in an *interaction_pair* for the *mechanical_system* to be a *mechanically_closed_system*.
- E42** `interaction_pair` := An *interaction_pair* is the form into which each *interaction_candidate* is placed in a *mechanically_closed_system*.
- E43** `mechanically_closed_system` := A *mechanically_closed_system* is a *mechanical_system* in which every *interaction_candidate* can be placed in an *interaction_pair*.

2.11 Non-law

- E44** `interaction_pair` := Two *interaction_candidates* form an *interaction_pair* when they correspond to the same *acting_object*, one being the *canonical_force*(*acting_object*) associated with that *acting_object* and the other being the *reaction_force*(*acting_object*) associated with that *acting_object*, the latter exerted on the *paired_particle*(*acting_object*).
- E45** `mechanically_closed_system` := A *mechanically_closed_system* is a *mechanical_system* whose future behaviour is determined by the equations produced from the interactions internal to it.

References

References

- [1] Daniel Kleppner and Robert Kolenkow, *An Introduction to Mechanics*, 2nd ed., Cambridge University Press, 2014.
- [2] John R. Taylor, *Classical Mechanics*, University Science Books, 2005.
- [3] Isaac Newton, *Philosophiæ Naturalis Principia Mathematica*.